

A. Data sources

Open Targets KG

Gene-disease associations
evidence

Target metadata

Interaction: PPI edges

NCBI data

B. Canonical analysis table

Canonical analysis table

One row per target-disease pair
~4.5M gene-disease pairs

F3: connected component
F4: biotype / type constraint



Factorial

16 filter combinations



H1

attention

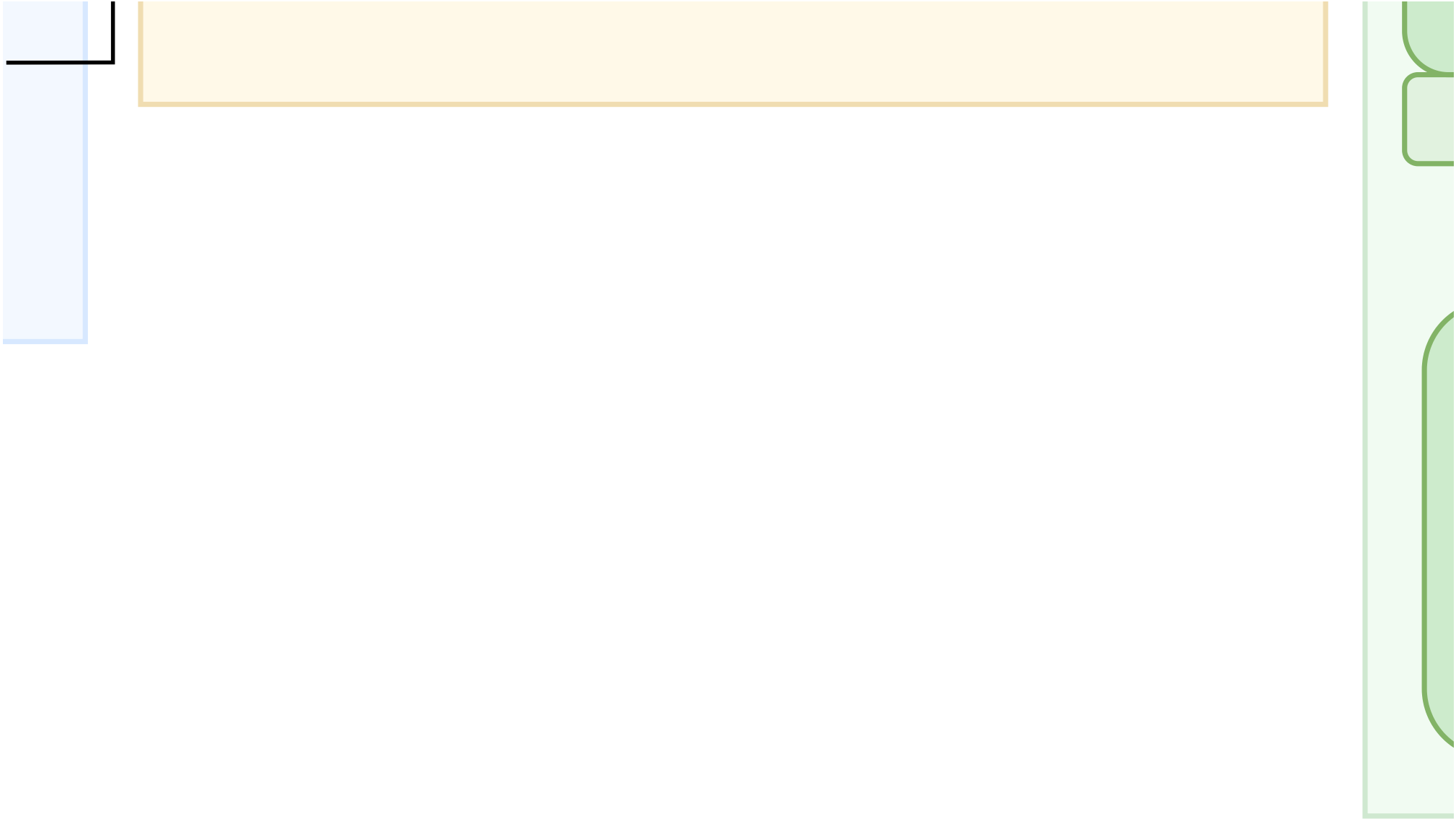
H2

nonlinear

H3

residual

gene2pubmed publication counts
per gene
gene2ensembl mapping



bias

compounding

structure

Hypothesis Tests



Interpretation

Filters compound non-linearly and systematically under-retain highly-studied pleiotropic hub genes.